MongoDB Operators

What are Operators in MongoDB?

- Operators are special keywords used to perform operations in queries
- They help filter, compare, and manipulate data
- Mostly used inside query objects
- Example: db.users.find({ age: { $gt: 18 } })

Types of MongoDB Operators

- Comparison Operators
- Logical Operators
- Element Operators
- Evaluation Operators
- Array Operators
- Update Operators

Comparison Operators

- $eq → Equal to
- $ne → Not equal
- $gt → Greater than
- $lt → Less than
- $gte → Greater than or equal
- $lte → Less than or equal

Example: db.users.find({ age: { $gt: 25 } })

Logical Operators

- $and → All conditions must be true
- $or → Any condition can be true
- $not → Negates a condition
- $nor → All conditions must be false

Example: db.users.find({ $or: [{age:{$lt:18}}, {age:{$gt:60}}] })

Element Operators

- $exists → Checks if a field exists
- $type → Checks the BSON data type of a field

Example: db.users.find({ phone: { $exists: true } })

Example: db.users.find({ age: { $type: 'int' } })

Evaluation Operators

- $regex → Pattern matching for strings
- $expr → Use aggregation expressions in queries
- $mod → Perform modulo operation

Example: db.users.find({ name: { $regex: '^A' } })

Example: db.users.find({ age: { $mod: [5,0] } })

Array Operators

- $all → Match arrays containing all elements
- $size → Match arrays with specific length
- $elemMatch → Match documents inside arrays
- $in → Check if array contains any value
- $nin → Opposite of $in

Example: db.posts.find({ tags: { $size: 3 } })

Example: db.posts.find({ tags: { $all: ['mongodb','database'] } })

Update Operators

- $set → Update field value
- $unset → Remove field
- $inc → Increase numeric value
- $push → Add value to array
- $inc → Increase or Decrease Number
- $mul → Number ko **multiply** karta hai
- $rename → Change field name
- $min → Field ko **minimum value se update** karta hai.
- $max → Field ko **maximum value se update** karta hai.

Array Update Operator:

- $push → Add element in array
- $addToSet → Duplicate allow nahi karta
- $pull → element **remove** karta hai

Example:
db.products.updateOne({name:'Laptop'},{$inc:{stock:10}})

Summary

- Operators make MongoDB queries powerful
- They allow filtering, updating and validating data
- Always start with $ symbol
- Examples: $gt, $or, $exists, $regex, $size

Thank You